

designed for Web development, and can be easily integrated with HTML. Hack supports fast development for PHP by enabling programmers to use both static and dynamic typing which, in turn, is called gradual typing.

Hack provides various important features, some of which are listed below.

Type annotations: This feature allows code to be explicitly typed on parameters, class member variables and return values.

Generics: This allows classes and methods to be parametrised in a manner similar to C# or Java.

Nullable types: This feature enables Hack to deal with nulls in a safer way by making use of the '?' operator.

Collections: This provides first class, built-in parametrised types such as Vector, Map, Set and Pair.

Lambdas: This allows the definition of first class functions.

Here's the syntax for the 'Hello World' program in Hack.

After installing HHVM, start it by typing the following:

```
hhvm -m server -p 8080
<?hh
Echo "Hello World!";
```

The advantages of Hack

- Hack code is mixed with PHP code. Instead of using `<?php` in PHP, use `<?hh` in Hack, which makes for a smooth migration between Hack and PHP.
- Hack runs with HHVM, which has a type-checking phase that verifies the consistency of the code.
- Hack is used to build complex websites at great speeds by ensuring proper organisation and error-free code, and gives programmers a unique safety advantage to write bug-free code.

Hack's disadvantages

- It contains abandoned features that make PHP a simple language.
- HTML code cannot be directly embedded into Hack and code can't be written outside of a function or class.
- Being a new language, Hack has lots of bugs and errors which, right now, make it an unstable language.

4. Rust

Rust is regarded as a general-purpose, multi-paradigm system programming language that is designed to meet three main objectives: safety, speed and concurrency. Rust was designed by Graydon Hoare from Mozilla Research, and uses LLVM as its backend. The design of the language has been refined due to the developer team's experience of writing the Servo Web browser layout engine and the Rust compiler. Rust, being an intelligent open source language, has an edge over other programming languages by having a number of compile-time safety checks, which generate no runtime overhead and eliminate all data traces. Rust has the inbuilt

functionality most needed for concurrent execution on multi-core machines, making concurrent programming memory safe without garbage collection. And it is the only language that does this job.

The latest stable release for Rust is 1.10.

The syntax for the 'Hello World' program in Rust is:

```
fn main() {
    println!("Hello, world!");
}
```

The advantages of Rust

- Rust is suitable for developers and projects where safety and stable execution, in addition to low-level optimisation and performance, are required. Rust adds lots of high-level functional programming techniques, making it feel like a low level and a high level language at the same time.
- Rust has an enlarged standard library, which is expanding constantly, focusing on file system access, networking, time and collections APIs.
- Rust is supportive of multi-platform development, ranging from Windows, Android and even ARM devices, apart from other platforms.

Rust's disadvantages

- Rust is a bit harder and more complex to learn and code. Error handling is a complex task, especially for newbies.
- Rust is a very immature language in terms of documentation, which is not available.
- Rust is currently not being adopted too much in the industry.

5. Scala

Scala (Scalable Language) is regarded as a general-purpose programming language designed to define programming patterns in a concise, elegant and type-safe way. Scala was developed by Martin Odersky in 2001 by using the Funnel programming language as the base. Scala basically integrates the features of object-oriented and functional languages, making programmers more productive and effective in writing source code. Scala has many of the features of functional programming languages like Scheme, Standard ML and Haskell, in addition to currying, type inference, immutability, lazy evaluation and pattern matching. Scala is more advanced compared to Java because it integrates other features like operator overloading, optional parameters, named parameters, raw strings and no checked exceptions. Scala and Java are related to each other by the fact that they are compiled to a bytecode and use the JVM. Scala is completely interoperable with Java.

The latest stable release of Scala is 2.11.8.